

ECE 306 HW 2

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1. 'display_line'
2. The character array representing the string ' NCSU \0' (NCSU surrounded by three spaces on either side, null terminated)
3. 0x008A8C
4. Pointer to 'char display[0]' is '0x00202C', pointer to 'char display[1]' is '0x00202E', pointer to 'char display[2]' is '0x002030', pointer to 'char display[3]' is '0x002032'
5. '0x00202C' contains a pointer to '0x002000', '0x00202E' contains a pointer to '0x00200B', '0x002030' contains a pointer to '0x002016', '0x002032' contains a pointer to '0x002021'
6. '[0x20, 0x20, 0x20, 0x4E, 0x43, 0x53, 0x55, 0x20, 0x20, 0x20, 0x00]'
7. $3.3\text{v} * 49.1\text{ mA} = 162.03\text{ mW}$ delivered $162.03\text{ mW} / 90\% = 180.03\text{ mW}$ supplied $180.03 / 2 = 90.02$ per battery Graph gives 1 AH for $\sim 90\text{ mW}$
 $1\text{ AH} / 0.0491\text{ A} = 20.37\text{ hours}$ $20.37\text{ hours} * 60 = 1222\text{ minutes}$ $1222\text{ minutes} / 20\text{ minutes/day} = 61.1\text{ days}$
8. It is located in '.bss' because it is uninitialized